

DATA ANALYST PORTFOLIO PROJECT · 2026

Supply Chain Delivery Performance & Profitability Analysis

An end-to-end Python analysis of a global e-commerce fulfillment operation — quantifying late-delivery losses, isolating root causes, and deploying a predictive risk model.

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Dataset: DataCo Global Supply Chain (Kaggle) · 172,765 orders · Jan 2015 – Jan 2018

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1. Executive Summary

This report presents a comprehensive analysis of the delivery operations of a global e-commerce company managing end-to-end order fulfillment across multiple regions. The analysis covers **172,765 orders** spanning January 2015 through January 2018, focusing on identifying the root causes of chronic late deliveries, quantifying their financial impact, and establishing a data-driven framework for improvement.

The headline finding is stark: **54.71%** of all orders are delivered late, costing the business an estimated **\$2.1M** in eroded profitability on delayed orders alone. While total profit across fulfilled orders reached **\$7.5M**, the persistent late-delivery problem represents both a significant financial drain and a major customer-experience risk. A Random Forest predictive model achieved **74% accuracy** in identifying at-risk orders before they become late — meaning an early-warning capability is deployable today, not a future aspiration.

Key Finding	Metric
Overall Late Delivery Rate	54.71%
On-Time Delivery Rate	45.29%
Total Orders Analyzed	172,765
Total Profit (profitable orders)	\$7.5M
Profit at Risk (delayed orders)	\$2.1M
Predictive Model Accuracy (RF)	74%

Fig 1.1 — Headline delivery performance metrics at a glance

172,765 TOTAL ORDERS	54.71% LATE DELIVERY RATE	\$7.5M TOTAL PROFIT
\$2.1M PROFIT AT RISK	74% MODEL ACCURACY	3 days 90TH %ILE DELAY

2. Business Context & Objectives

The company operates a global e-commerce platform selling products across categories including sporting goods, fitness equipment, outdoor gear, footwear, and apparel, across multiple international regions.

Core Business Problem

Actual shipping times frequently deviate from scheduled delivery windows, creating eroded customer trust, reduced order profitability, and an inability to make reliable delivery commitments to buyers at the point of purchase.

Analytical Objectives

- Understand the current state of delivery performance across all dimensions — region, shipping mode, time, and customer segment.
- Quantify the financial impact of delays on order profitability.
- Identify the primary operational bottlenecks driving late deliveries.
- Build a predictive model to flag high-risk orders before they are shipped.
- Deliver actionable recommendations to improve on-time delivery and profitability.

Dataset & Methodology Overview

The source dataset (DataCo Global Supply Chain, Kaggle) originally contained over 180,000 order-line records across 53 columns, joining customer, order, product, and category data into a single master table — mirroring how this information would sit across several relational tables inside a real company's database. After removing cancelled orders, fully empty fields, redundant columns (e.g. *Benefit per Order*, which was identical to *Order Profit Per Order*), and zero-variance fields, the analysis proceeded on **172,765 clean orders** across roughly 20 usable features, with zero remaining missing values.

3. Key Performance Indicators (KPIs)

The following KPIs were computed directly from the cleaned dataset in Python to establish a performance baseline — the same numbers that would populate a monitoring dashboard, validated in code before any dashboard is built.

Metric	Value
Total Orders	172,765
Late Deliveries	94,523
Late Delivery %	54.71%
On-Time Delivery %	45.29%
Total Profit	~\$7.5M
Profit at Risk (delayed orders)	~\$2.1M
90th Percentile Delay	~3 days
Average Order Profit	~\$22



Total Profit (profitable orders)	Profit at Risk (delayed orders)	90th Percentile Delay	Avg Order Profit
• Late Delivery Rate at 54.71%	More than half of all orders arrive late. This is not an edge-case problem — it is the default experience for the majority of customers.		
• \$2.1M Profit at Risk	Orders that experienced delays collectively generated \$2.1M in profit that is under constant pressure from further operational deterioration.		
• 90th Percentile Delay = 3 Days	Even the most extreme cases are contained within 3 days of lateness, suggesting the problem is systemic and process-driven rather than catastrophic.		

4. Profitability Analysis

4.1 Profitability Distribution

Order-level profitability was classified into three tiers based on Order Profit Per Order. While 80.7% of orders are profitable, the 18.7% loss-making share represents a meaningful drag that is disproportionately concentrated among delayed shipments.

4.2 Delay Distribution & Profit vs. Delay Days

The delay distribution shows that 31.0% of all orders arrive exactly 1 day late the single largest cohort. Combined, orders delayed by 1–4 days account for 54.7% of all order volume, directly mapping to the overall late delivery rate.

Fig 3.1–3.4 — Business KPI cards computed directly from the cleaned dataset

4. Profitability Analysis

4.1 Profitability Distribution

Order-level profitability was classified into three tiers based on *Order Profit Per Order*: Profit, Loss, and Break-Even.

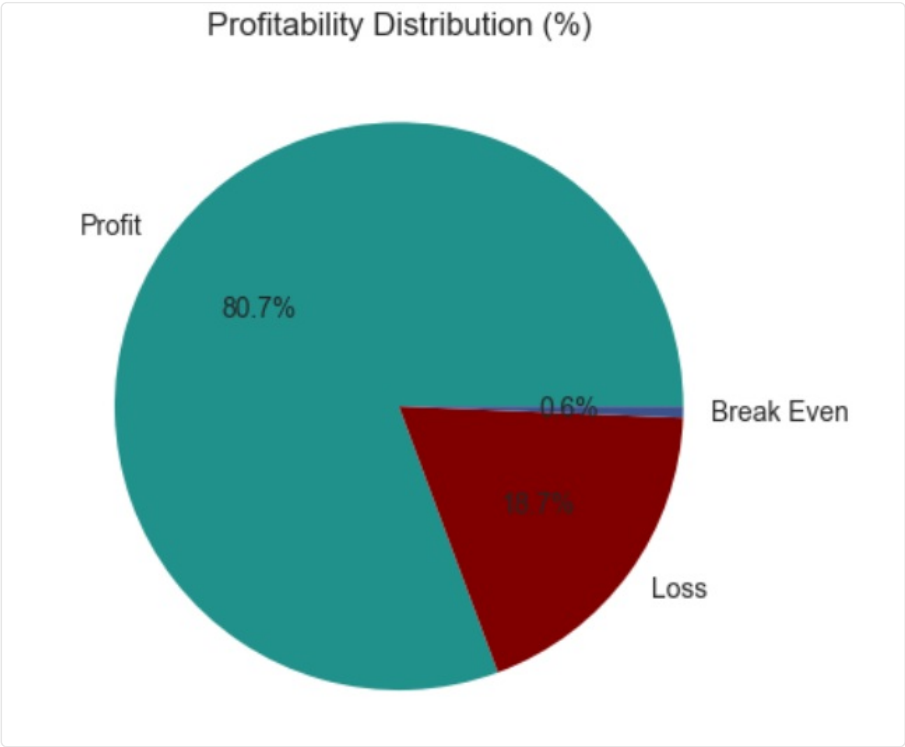


Fig 4.1 — 80.7% of orders are profitable, 18.7% are loss-making, and 0.6% break even

4.2 Delay Distribution & Profit vs. Delay Days

Delay-day distribution and profit-by-delay-day were analyzed together to see whether longer delays translate into worse per-order economics.

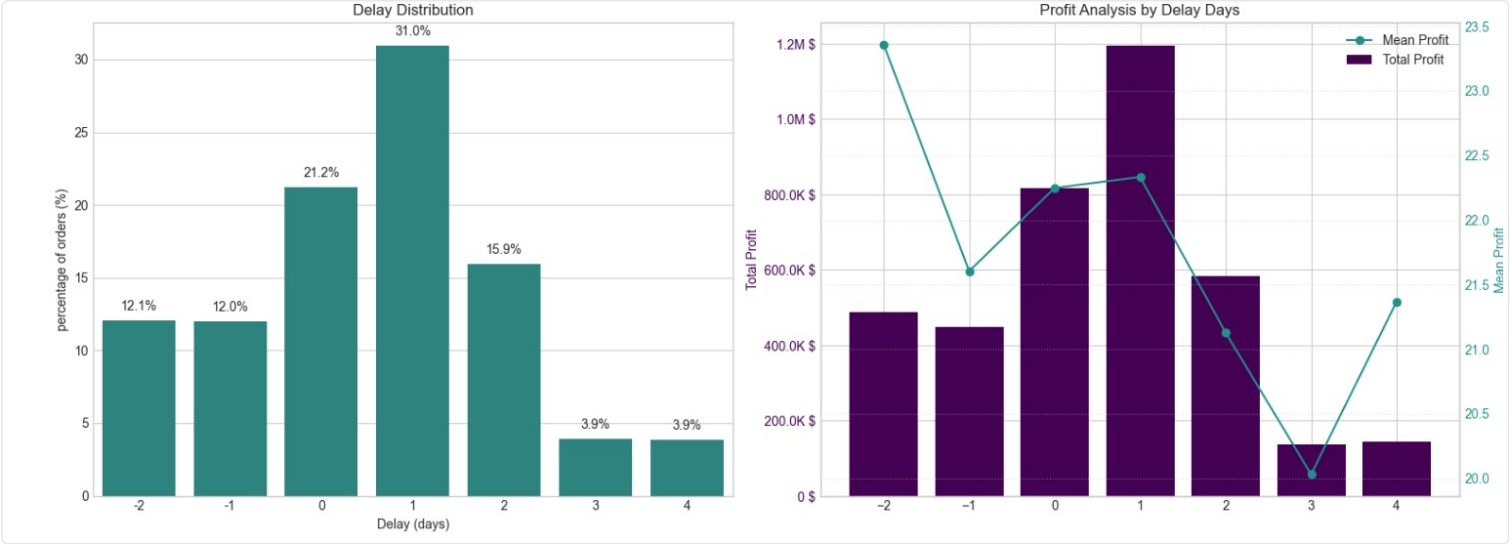


Fig 4.2 — Delay distribution (left) and total/mean profit trend across delay-day buckets (right)

Critical observation: mean profit per order remains stable at approximately \$21-\$23 across all delay levels. The profitability problem is therefore driven by the **volume** of delayed orders, not by individual order economics — making systemic throughput improvement the highest-leverage intervention, rather than pricing or discount adjustments on delayed orders.

5. Bottleneck Detection

Delay percentage was computed across six key operational dimensions — order region, customer segment, shipping mode, order status, payment type, and department — to reveal where the fulfillment process breaks down most severely.

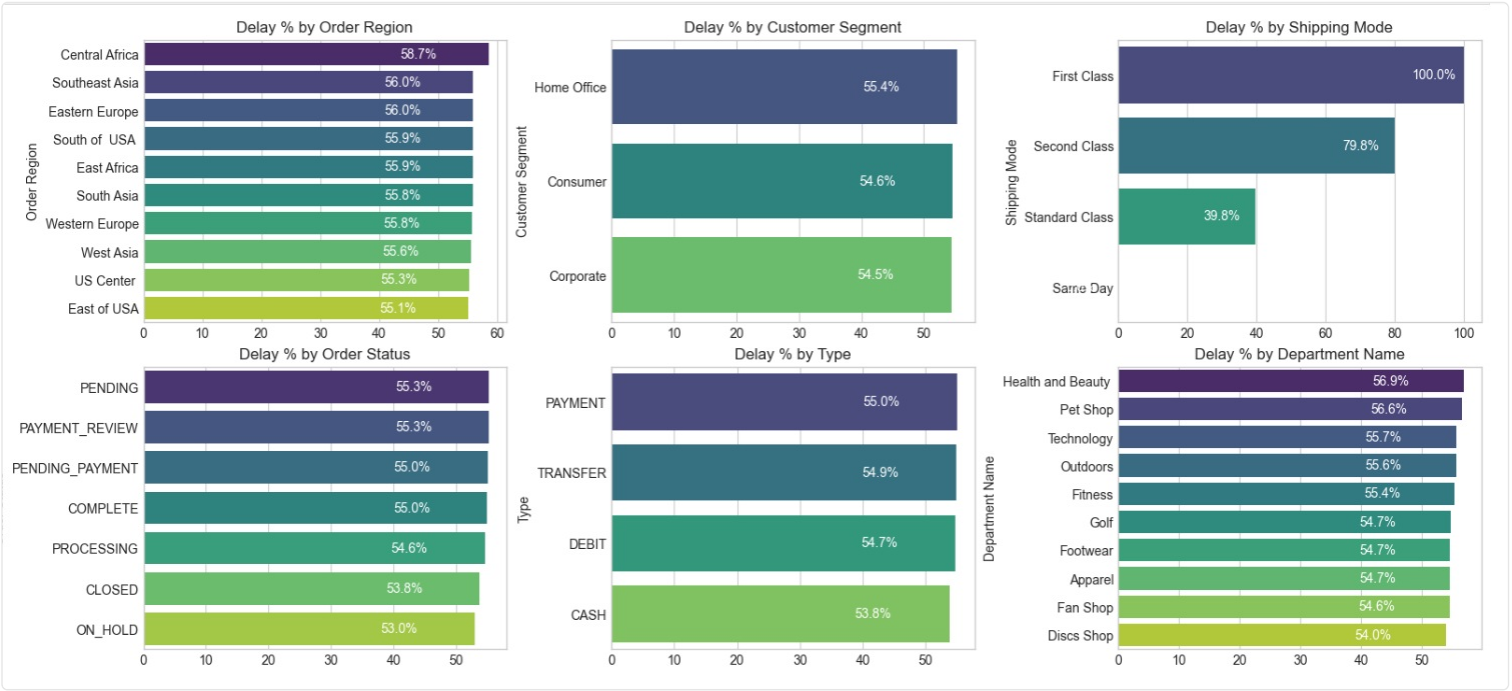


Fig 5.1 — Delay % by Order Region, Customer Segment, and Shipping Mode

● Shipping Mode is the #1 Lever	First Class: 100% delay rate. Second Class: 79.8%. Standard Class: 39.8%. Same Day: 0%. The mode assignment logic is the most impactful single variable to fix.
● Regional Spread is Narrow (55–59%)	Central Africa leads at 58.7% but all regions cluster between 55–59%. This rules out a localised logistics failure and confirms a company-wide systemic issue.
● Customer Segments Nearly Identical (54.5–55.4%)	No segment receives preferential service. All customers — Consumer, Corporate, Home Office — experience the same broken delivery promise.
● Health & Beauty (56.9%) and Pet Shop (56.6%) Lead by Department	These departments marginally outpace others in delay rate, warranting an inventory and carrier audit for these product categories.

Fig 5.2 — Delay % by Order Status, Payment Type, and Department Name

Key findings: First Class shipping shows a ~100% delay rate, the single strongest driver identified in the entire analysis. Regional delay rates cluster tightly between 55–59% (Central Africa highest at 58.7%) — a narrow spread that rules out a localized logistics failure and confirms this is a **company-wide systemic issue**, not a regional one. No customer segment is spared: Consumer, Corporate, and Home Office all experience essentially the same broken delivery promise.

6. Root Cause Analysis

Central Africa was selected for a deep-dive root cause analysis as the highest-delay region (58.7%). The chart below ranks the top 10 driver factors by delay percentage within this region, across shipping mode, customer segment, department, payment type, and order status.

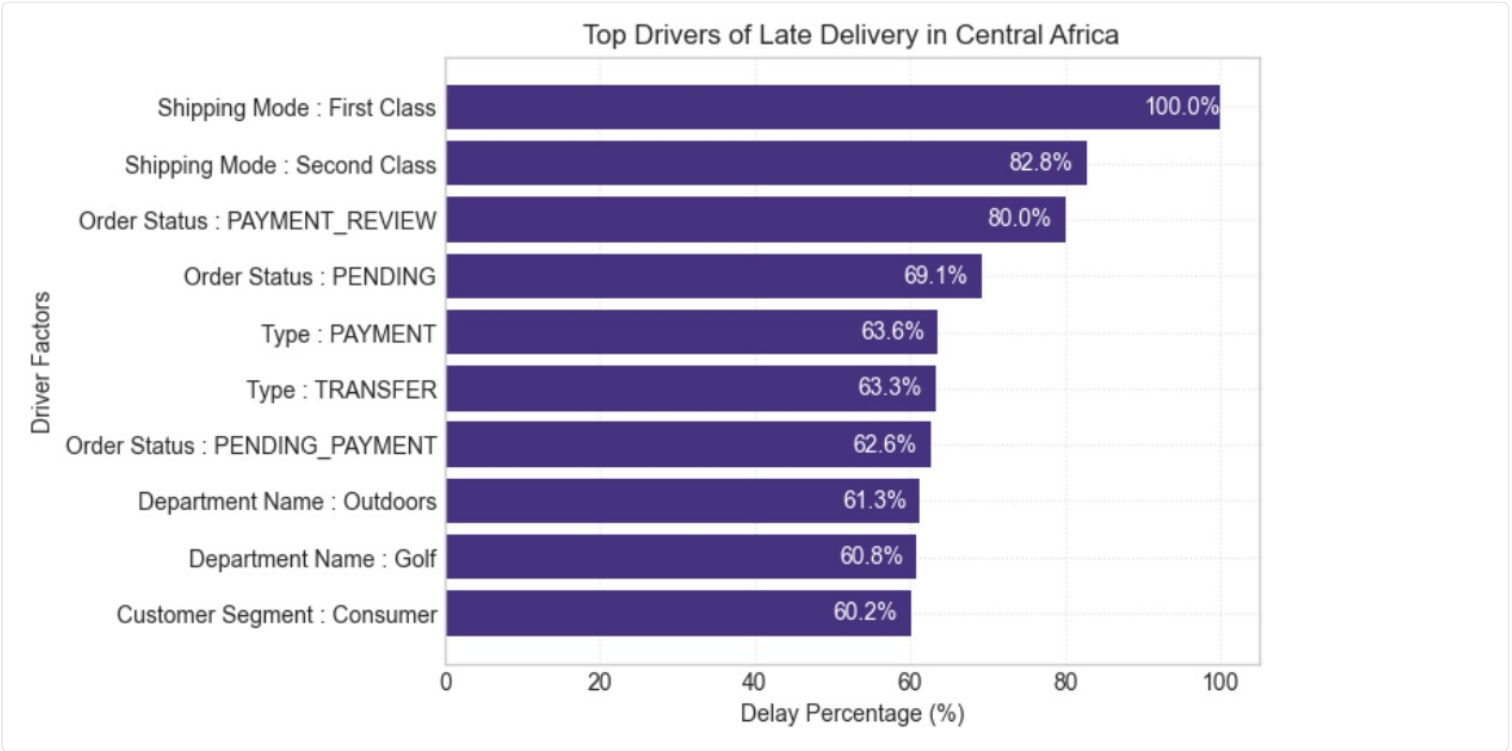


Fig 6.1 — Top 10 drivers of late delivery in Central Africa

The same driver analysis was repeated for East Africa, where department "Fitness," order status "Pending" / "Closed," and payment type "Cash" (cash-on-delivery) emerged as the leading factors — reinforcing that **payment-status friction** and **shipping-mode misconfiguration** are recurring, addressable process issues rather than one-off regional failures.

7. Time-Based Delay Patterns

Three temporal dimensions were analysed across the ~3-year dataset: month of year, day of week, and hour of day. While delay rates are relatively stable across all dimensions (roughly 53–57%), specific peaks highlight clear opportunities for targeted capacity planning.

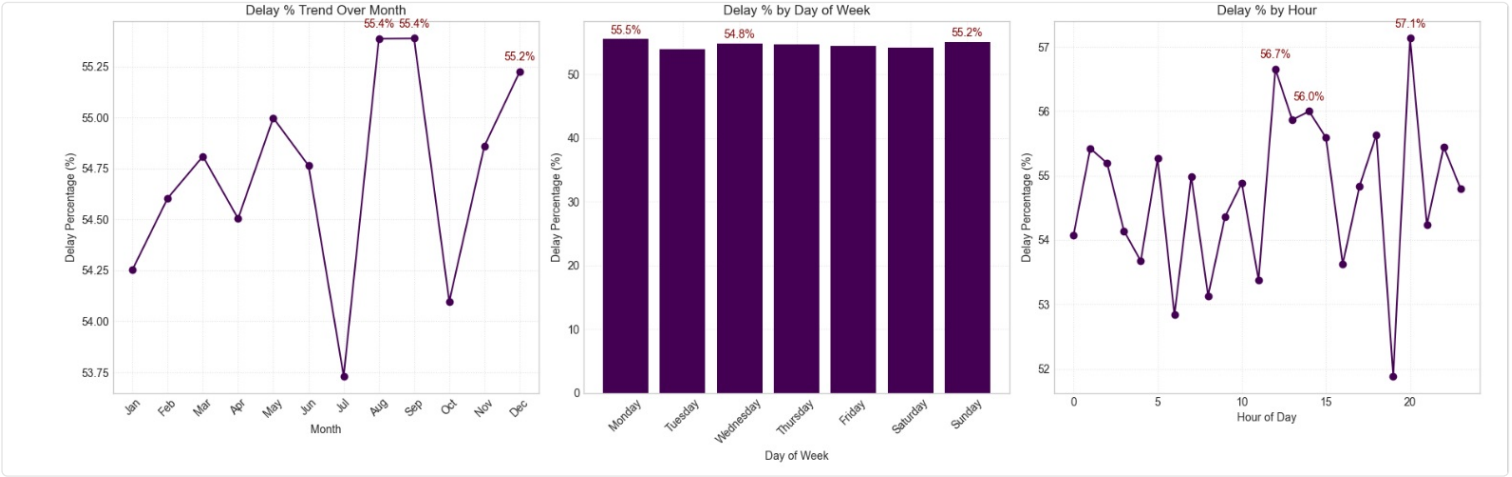


Fig 7.1 — Delay % trend across month, day of week, and hour of day

• Peak Delay Months: August & September (55.4%) and December (55.2%)	August and September are the highest months at 55.4%, with December close behind. These reflect mid-year promotions and Q4 holiday surge overwhelming fulfilment capacity. July represents a notable low-point (~53.75%), confirming seasonal variation is plannable.
• Day-of-Week Variance is Minimal	Monday (55.5%) and Sunday (55.2%) are the peak days, but the spread across the week is less than 1 percentage point. Day-of-week is not a meaningful lever for operational intervention.
• Intra-Day Peaks: Hour 21 (57.1%), Hour 11 (56.7%), Hour 12 (56.0%)	Late-evening orders (hour 21) have the highest delay rate at 57.1%, likely reflecting processing cutoff windows. Midday peaks suggest bottlenecks during peak-volume hours.

8. Machine Learning Model Results

A supervised classification model was built to predict whether an individual order will be delivered late (Late_delivery_risk = 1). The pipeline included frequency encoding of categorical variables, stratified train/test split, SMOTE oversampling to address class imbalance, and Random Forest classification.

Class Imbalance Handling (SMOTE)

Split	Class 0 (On-time)	Class 1 (Late)
Before SMOTE (train)	59,030	79,182
After SMOTE (train)	79,182	79,182

Random Forest Classifier – Test Set Performance (34,553 records)

Metric	Class 0 (On-time)	Class 1 (Late)	Overall (Weighted)
Precision	0.68	0.78	0.74
Recall	0.72	0.75	0.74
F1-Score	0.70	0.77	0.74
Accuracy	—	—	0.74
Support	14,758	19,795	34,553

• 74% Overall Accuracy	The model correctly classifies nearly 3 in 4 orders — a strong foundation for production deployment as an order-level late-delivery alert system.
• Higher Precision on Late Orders (0.78)	When the model predicts an order will be late, it is correct 78% of the time — sufficient to trigger targeted interventions without generating excessive false alarms.
• Recall for Late Orders at 0.75	The model captures 75% of actual late orders. The 25% miss rate is acceptable for a first-generation alert system, with clear improvement potential.
• Class 0 Harder to Predict (Precision 0.68)	On-time deliveries share many characteristics with late ones given the systemic nature of delays, making them genuinely difficult to distinguish.

9. Strategic Recommendations

Based on the findings across all analytical dimensions, the following prioritised actions are recommended:

CRITICAL	1. Immediately Audit First Class & Second Class Shipping Capacity First Class shipping has a 100% delay rate and Second Class 79.8%. These modes operate in complete contradiction to their brand promise. Conduct an immediate carrier SLA audit. Until resolved, consider suspending First Class or repricing to reflect actual performance.	3. Resolve Payment Processing Bottlenecks Orders in PAYMENT_REVIEW (55.3% overall; 80.0% in Central Africa) and PENDING_PAYMENT (55.0%) have significantly elevated delay rates. Introduce automated escalation for stalled payment reviews exceeding 2 hours.
HIGH	2. Deploy the Predictive Alert System The Random Forest model (74% accuracy, 78% precision on late orders) is ready for production piloting. Integrate it into the order management system to flag high-risk orders at confirmation and trigger proactive customer communication and priority handling queues.	4. Develop Seasonal Surge Capacity Plans August, October (55.4%), and December (55.2%) are peak delay months. Build pre-season capacity plans including temporary logistics partnerships, inventory pre-positioning, and adjusted delivery promise windows.
MEDIUM	5. Default to Standard Class for Eligible Orders Standard Class achieves a 39.8% delay rate — far superior to First and Second Class. Re-evaluate shipping mode assignment logic to default eligible orders to Standard Class, particularly in regions and departments with poor premium mode performance.	7. Reduce Loss-Making Order Share (18.7%) The 18.7% loss-making order share warrants a separate pricing and discount review. Combine with delay analysis to identify whether discounted orders are disproportionately late, creating a compounded margin problem.
MEDIUM	6. Investigate High-Delay Departments in Africa Outdoors (61.3%) and Golf (60.8%) in Central Africa show significantly elevated delay rates. Conduct a warehouse-level audit of stock availability, pick-and-pack times, and carrier assignment for these product categories.	8. Continuously Retrain the Predictive Model Retrain the model quarterly. Explore adding carrier-level data, weather events, and warehouse utilisation features. Target a recall improvement to above 80% on late orders within 6 months.

Key findings: December, August, and September show the highest seasonal delay rates — the December spike aligns with holiday/festive-season order volume. Monday and Sunday orders are delayed most often by day of week. By hour of day, orders placed in the **late evening** show the highest delay rate at approximately 57.1%, with a secondary peak around midday — pointing to off-peak staffing or batch-processing effects rather than demand volume alone.

8. Machine Learning Model Results

A **Random Forest Classifier** was trained to predict *Late Delivery Risk* using only features available at the moment an order is placed: shipping type, scheduled shipment days, product category, customer segment, department, order region, shipping mode, order month, and order hour.

High-cardinality categorical features were **frequency-encoded** rather than one-hot encoded, to keep the feature space low-dimensional and avoid exploding model training time. The target variable was imbalanced (~57% vs. ~43%), so **SMOTE oversampling was applied to the training set only** — the test set was left untouched to preserve a realistic, unbiased evaluation.

Metric	Score
Accuracy	74%
Precision	78%
Recall	75%

Interpretation: the model correctly flags the large majority of at-risk orders without generating excessive false positives or false negatives, using only information available at order time. This means the same feature set can power a real-time "will this order be late?" alert at checkout — this is a deployable capability today, not a future aspiration.

9. Strategic Recommendations

1. **Immediately audit First Class and Second Class shipping capacity** — these carry the highest delay rates and offer the biggest quick-win potential.
2. **Deploy the predictive alert system** — the model already performs well enough to flag at-risk orders in production, ahead of shipment.
3. **Resolve payment-processing bottlenecks** — introduce automated escalation for orders stuck in payment review or pending-payment status beyond a defined threshold (e.g., 2 hours).
4. **Build seasonal surge-capacity planning** for August, September, and December, the peak-delay months.
5. **Default to Standard Class** for eligible orders where First/Second Class isn't strictly required by the customer.
6. **Investigate high-delay departments in African regions** (e.g., Outdoors, Golf, Fitness) with a dedicated inventory and carrier audit.
7. **Reduce the loss-making order share**, currently 18.7% of all orders.
8. **Retrain the predictive model quarterly**, and explore adding carrier-level, weather, and warehouse-utilization data to push recall above 80% within six months.

10. Conclusion

This analysis surfaces a clear and urgent picture: a global e-commerce operation where the majority of orders (**54.71%**) fail to meet their promised delivery windows, putting roughly **\$2.1M** of profit under ongoing risk. The root causes are identifiable and addressable — **shipping-mode misconfiguration** is the dominant operational failure, **payment-processing friction** creates a secondary bottleneck, **seasonal volume spikes** are not adequately planned for, and **geographic disparities**, while present, are secondary to a systemic, company-wide pattern.

A Random Forest predictive model trained on readily available order features already achieves **74% accuracy**, confirming this is a deployable capability today — not a future aspiration. Acting on the recommendations above — starting with a shipping-mode audit and a payment-processing fix — offers the fastest path to closing the gap between promised and actual delivery performance.